

MIRACLE: An Online, Explainable Multimodal Interactive Concept Learning System

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Abstract

We present *MIRACLE*, a system for online, interpretable visual concept and video action recognition. Through a chat interface, users query the recognition system with an uploaded image or video. For images, *MIRACLE* returns concept predictions from its structured knowledge base, justifying its predictions with heatmaps and natural language-based attribute detections. For videos, *MIRACLE* predicts an action and justifies its prediction with time varying entity-entity relations. With its ability to learn new concepts in an online, few-shot manner and its support of dynamic changes to its knowledge base, *MIRACLE* represents a step forward in interpretable multimodal learning systems.

CCS Concepts

• **Computing methodologies** → **Visual inspection; Activity recognition and understanding; Video summarization; Discourse, dialogue and pragmatics.**

Keywords

Multimodal Interaction, Online Learning, Object Recognition, Video Action Detection, Few-shot Learning, Interpretability

ACM Reference Format:

Ansel Blume, Khanh Duy Nguyen, Zhenhailong Wang, Yangyi Chen, Michal Shlapentokh-Rothman, Xiaomeng Jin, Jeonghwan Kim, Zhen Zhu, Jiateng Liu, Kuan-Hao Huang, Mankeerat Sidhu, Xuanming Zhang, Vivian Liu, Raunak Sinha, Te-Lin Wu, Abhay Zala, Elias Stengel-Eskin, Da Yin, Yao Xiao, Utkarsh Mall, Zhou Yu, Kai-Wei Chang, Camille Cobb, Karrie Karahalios, Lydia Chilton, Mohit Bansal, Nanyun Peng, Carl Vondrick, Derek Hoiem, and Heng Ji. 2024. *MIRACLE: An Online, Explainable Multimodal Interactive Concept Learning System*. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*, October 28–November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3664647.3684993>

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MM '24, October 28–November 1, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3684993>

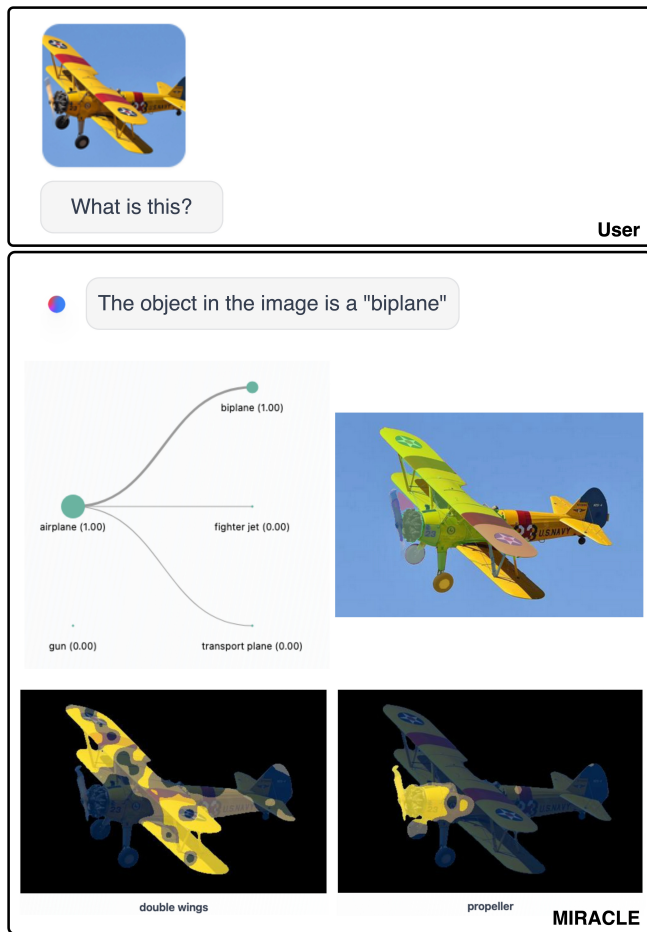


Figure 1: Example of image recognition. MIRACLE returns its predicted concept, the prediction’s reasoning path through the knowledge base, and the prediction’s explanation.

1 Introduction

Modern vision systems have achieved tremendous breakthroughs, with some boasting strong performance in multiple core vision tasks, while others excel at one task, approaching human performance in a single setting [1, 4, 10]. Multimodal frameworks, in particular, are able to utilize natural language supervision in order to introduce flexibility and zero-shot transfer to downstream tasks [6, 7, 12]. However, these models are black boxes—they produce outputs with no justification for their predictions. In critical infrastructure, it is necessary to have confidence that a system not only achieves high quantitative performance, but also makes its decisions for the right reasons—and not, for example, due to artifacts in the model’s training data. An ideal system would explain its predictions both in terms of linguistically grounded visual attributes, and in terms of non-textual perceptual features. Structured knowledge representations also assist with interpretability, as organizing a system’s learned concepts facilitates external audits and encourages transfer between similar classes. Unfortunately, standard visual

classification paradigms, e.g. using a classification head on an image backbone, use difficult-to-interpret neural networks without structured representations, making such paradigms unfavorable for interpretability.

Such models also lack the dynamism of human concept learning. Most struggle to learn new concepts on the fly from just a few examples, and the addition of new concepts may require retraining all classes. Few-shot frameworks, while capable of impressive generalization from few examples, typically utilize uninterpretable representations, or must be re-optimized from scratch to accommodate new classes [2, 13, 15]. An ideal online learning system should retain the ability to perform well from minimal examples, improve as more data is provided, and learn new concepts without retraining the system from scratch.

To address these deficiencies we present *MIRACLE*, an online explainable Multimodal InteRActive Concept LEarning System. MIRACLE learns concepts from images in an online, few-shot manner. Its multimodal concept representations, built upon CLIP [12], DINOv2 [10], and SAM [5], are intrinsically interpretable, with MIRACLE justifying its image predictions via natural-language attributes and non-textual visual features. MIRACLE’s concept knowledge base is represented as a directed acyclic graph, with concept nodes connected by both instance (*is a*) and part (*has a*) relations. MIRACLE also performs video action recognition, identifying key frames and entity-entity relations that support and validate its predictions. Users interact with the system via an intuitive chat-based interface which interprets both templated instructions and free-form commands. With its interpretable predictions, structured knowledge base, and dynamic learning capabilities, MIRACLE represents a step forward in interpretable multimodal learning systems.

2 Visual Concept Recognition

The first MIRACLE subsystem is for visual concept recognition (Figure 1). The concept recognition system learns a linear model over a collection of multimodal features for each concept. The linear model enables feature importance analysis via the sign and magnitude of the model’s weights. Image features are interpretable by design, and consist of: 1) a global visual score computed from frozen DINOv2 image features [10] which are refined by region pooling [14] over the foreground predicted by Rembg [3]; 2) concept-specific textual attribute scores predicted by CLIP [9, 12] (e.g. a fork has *tines* and a *handle*); 3) general attribute scores using predictors trained on VAW [11] (e.g. *color*, *shape*, *material*). Each of these scores are computed for the entire image, and for crops of image regions generated by SAM [5] for better grounding of the prediction in image regions. These scores, along with key image regions, are returned to the user to explain the system’s prediction. The combination of zero-shot attribute scores and learnable components enables strong few-shot generalization: with as little as a concept name and three exemplar images, MIRACLE learns to distinguish the concept from others in its knowledge base.

MIRACLE’s knowledge base is also designed to be readily interpretable. Concepts are organized as a DAG with both instance (*is-a*) and part (*has-a*) edges. For example, a biplane *is an* airplane, but *has a* set of double wings. This graphical representation enables hierarchical prediction, where MIRACLE can confidently predict an

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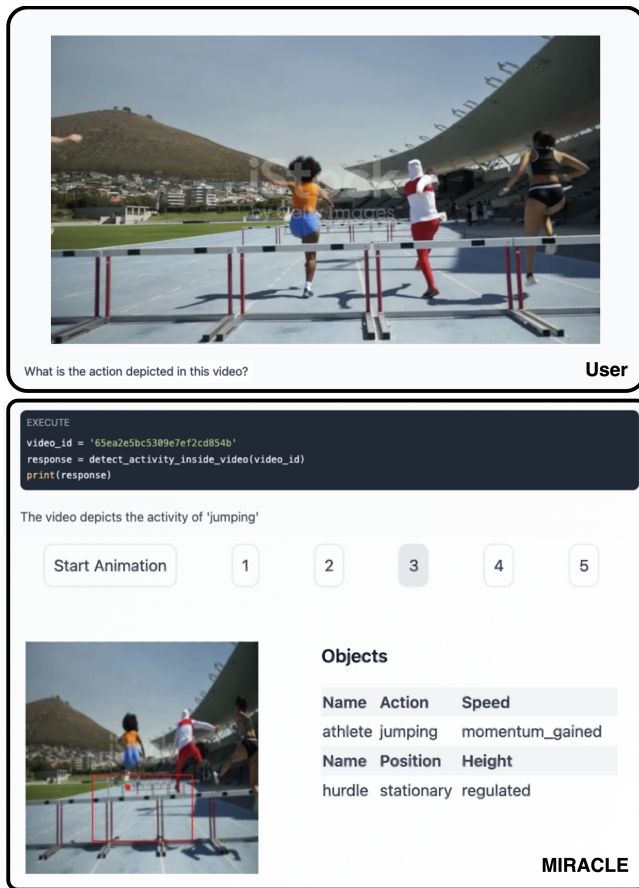


Figure 2: Example of video action detection. MIRACLE predicts an action, explaining the prediction with keyframes annotated with bounding boxes and entity-entity relations.

image’s general class (airplane) before attempting to determine its fine-grained one (biplane). Part predictions are incorporated into the whole concept’s prediction score—if we detect the *double wings* part, it is more likely that a *biplane* is in the image.

3 Video Action Recognition

MIRACLE also has a video action recognition subsystem. This subsystem consists of a Paxion model [16] trained to detect a fixed set of actions, and LLaVA [8] finetuned to interpret Paxion’s predictions. Specifically, we use Paxion to compute the video’s keyframes and regions with high attention weights, and annotate these regions with bounding boxes. The LLaVA model then uses these annotated keyframes alongside Paxion’s prediction to explain the prediction in terms of detected entities and relations in the frames (Figure 2).

4 Live Interactive Demonstration

For our live demo, we will have stations set up for individuals to interact with MIRACLE. Each station will have a set of prepared images and videos which can be uploaded to observe system outputs. Users will be encouraged to upload their own images and extend

MIRACLE’s knowledge base. A pre-recorded demonstration will be available for viewers to obtain a high-level understanding without needing hands-on interaction.

5 Acknowledgments

This research is based upon work supported by U.S. DARPA ECOLE Program No. HR00112390060. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies, either expressed or implied, of DARPA, or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright annotation therein.

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